# Time and Space Complexity of the CAD Shape Descriptors

Analysis of the 15-descriptor pipeline, with measured scaling

QRCAD: training-free CAD shape retrieval

Companion to *CAD_Descriptors_Explained.pdf*

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This document gives the time and space complexity of every step in the descriptor pipeline: mesh preprocessing, the fifteen descriptors exactly as defined in the companion document, and the assembly, compression and search stages. Each bound is derived line by line from the algorithm and then checked against measurements. The kernels were implemented in NumPy/SciPy and timed over sizes spanning one to two orders of magnitude, and the log–log slope of each curve was compared with the predicted exponent.

Measured times come from a 2-core cloud container (single process, NumPy 2.4, SciPy 1.17). They show scaling and relative cost; absolute times on a workstation will be smaller.

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# 1 Cost model and notation

## 1.1 What is being measured

**Time** is the number of arithmetic operations and memory accesses, counted in the word-RAM model: each floating-point operation, comparison or array access costs $\mathcal{O}(1)$. **Space** is the working memory needed *in addition to* the input mesh, which itself occupies $\Theta(F)$ words. "Expected" refers to randomised steps (hashing, random sampling, BVH traversal on typical geometry). "Worst case" is stated separately where it differs.

Two kinds of space matter in practice, and both are reported:

- **Vectorised space**: what a NumPy implementation uses when it builds whole arrays at once, e.g. all $S$ random quadruples as an $S \times 4 \times 3$ array. This is fastest in Python but memory grows with $S$.
- **Streamed space**: what the same algorithm needs if the samples are processed in fixed-size chunks and only the histogram is kept. This is the true lower bound of the algorithm.

## 1.2 Symbols and default values

All defaults below are the values used in *CAD_Descriptors_Explained.pdf*. The mesh size is a typical exported CAD STL.

| Symbol | Meaning | Default |
|---|---|---|
| $F$ | triangles in the mesh | $5 \times 10^4$ |
| $V$ | welded vertices; for a closed mesh $V = \frac{1}{2}F + 2 - 2g \approx F/2$ | $2.5 \times 10^4$ |
| $E$ | edges; for a closed mesh $2E = 3F$ | $7.5 \times 10^4$ |
| $n$ | surface sample pool | $2 \times 10^4$ ($2 \times 10^5$ for CRS-A) |
| $S$ | random tuples (pairs, triples, quadruples) | $2 \times 10^5$ |
| $B$ | histogram bins | 32 (36, 64, 12×8 where stated) |
| $h$ | vertices on the convex hull of the pool | $\leq n$ |
| $L$ | spherical-harmonic band limit | 16 |
| $K$ | Laplace–Beltrami eigenvalues | 30 |
| $m$ | Lanczos (Krylov) subspace size, $\max(2K + 1, 20)$ | 61 |
| $k$ | neighbours per point for curvature | 30 |
| $N_\varphi$ | angular bins in CRS-A | 256 |
| $N_{\text{fam}}$ | planar face families found by PAF (capped) | $\leq 64$ |
| $c$ | rays per sample in MTS | 16 |
| $n_s$ | simplices in the filtration, $V + E + F \approx 3F$ | $1.5 \times 10^5$ |
| $M$ | models in the database | 600 |
| $d$ | final descriptor length | 781 |

## 1.3 Mesh identities used throughout

For a closed, orientable, manifold triangle mesh of genus $g$, Euler's formula $V - E + F = 2 - 2g$ together with $2E = 3F$ gives

$$E = \tfrac{3}{2}F, \qquad V = \tfrac{1}{2}F + 2 - 2g. \tag{1}$$

Bounds in $V$, $E$ or $F$ are therefore interchangeable up to constant factors, and the analysis uses whichever is most natural. Every vertex has average degree $2E/V \approx 6$, so the cotangent Laplacian has about $7V$ non-zeros.

### 1.4 How the measurements were made

Each kernel from the companion document was implemented in vectorised NumPy/SciPy, applied to synthetic inputs of increasing size, and timed as the best of two runs. The inputs are closed torus meshes with $F$ from $1.25 \times 10^4$ to $4 \times 10^5$, subdivided box meshes for PAF, and area-weighted surface samples. Peak memory was measured with Python's `tracemalloc`, which records NumPy allocations but not memory allocated inside compiled libraries (SuperLU, ARPACK). For LBE the size of the sparse LU factor is reported instead. The *measured exponent* is the least-squares slope of $\log t$ against $\log(\text{size})$. PHD was measured for its $H_0$ union–find core only, and MTS was not measured, because the libraries they need (GUDHI and a BVH ray tracer) are not available in the sandbox. Both are analysed in full.

## 2 Summary of results

### 2.1 Complexity of every step

Space is given as *streamed / vectorised* where the two differ (see §1.1). "Ops" is the approximate number of arithmetic operations at the default sizes. "Measured" is the wall-clock time at the default sizes on the test container, and "exp." is the fitted log–log exponent, with the predicted value in brackets.

| Step | Time | Space | Ops | Measured | exp. (pred.) |
|---|---|---|---|---|---|
| **Shared preprocessing (once per model)** | | | | | |
| P1 face data | $\Theta(F)$ | $\Theta(F)$ | $1.5\times10^6$ | | with P3 |
| P2 weld | $\mathcal{O}(F)$ hash; $\mathcal{O}(F\log F)$ sort | $\Theta(F)$ | $3\times10^6$ | 0.137 s | 1.09 (1) |
| P3 edge map | $\mathcal{O}(F)$ hash; $\mathcal{O}(F\log F)$ sort | $\Theta(F)$ | $2\times10^6$ | 0.0161 s | 1.14 (1) |
| P4 sampling | $\mathcal{O}(F + n\log F)$ | $\Theta(F+n)$ | $3\times10^5$ | 0.00512 s | 0.97 (1) |
| P5 diameter, iteration | $\mathcal{O}(tn)$ | $\Theta(n)$ | $10^5$ | 0.003 s | 1.00 (1) |
| (exact hull) | $\mathcal{O}(n\log n + h^2)$, up to $\mathcal{O}(n^2)$ | $\Theta(n)$ | $10^8$ | 5.22 s | 2.00 (2) |
| P6 principal axes | $\Theta(n)$ | $\mathcal{O}(1)$ | $2\times10^5$ | <1 ms | – |
| P7 kd-tree | $\mathcal{O}(n\log n)$ | $\Theta(n)$ | $3\times10^5$ | | in curvature |
| P8 BVH | $\mathcal{O}(F\log F)$ | $\Theta(F)$ | $10^6$ | | not measured |
| **Descriptors** | | | | | |
| D1 | $\Theta(F+S)$ | $\Theta(B)$ / $\Theta(S)$ | $2\times10^6$ | 0.0107 s | 0.95 (1) |
| D2 | $\Theta(S)$ | $\Theta(B)$ / $\Theta(S)$ | $2.4\times10^6$ | 0.0122 s | 0.98 (1) |
| Triangle | $\Theta(S)$ | $\Theta(B)$ / $\Theta(S)$ | $9\times10^6$ | 0.0281 s | 1.07 (1) |
| D4 | $\Theta(S)$ | $\Theta(B)$ / $\Theta(S)$ | $1.6\times10^7$ | 0.121 s | 1.08 (1) |
| SHPS | $\Theta(FL^2)$ | $\Theta(F+L^2)$ | $4.6\times10^7$ | 0.0528 s | 1.10 (1) |
| NCF | $\Theta(S)$ | $\Theta(B)$ / $\Theta(S)$ | $3\times10^6$ | 0.0205 s | 0.99 (1) |
| LBE | $\mathcal{O}(V^{3/2} + rm(\text{nnz}_{LU} + mV))$ | $\mathcal{O}(\text{nnz}_{LU} + mV)$ | $10^8$–$10^9$ | 1.05 s | 1.28 (1–1.5) |
| DAD | $\Theta(F)$ | $\mathcal{O}(1)$ / $\Theta(F)$ | $2.3\times10^6$ | 0.0136 s | 1.05 (1) |
| Curvature | $\mathcal{O}(n\log n + nk)$ | $\Theta(n)$ / $\Theta(nk)$ | $3\times10^7$ | 0.26 s | 1.09 (1+) |
| CRS-R | $\Theta(n)$ | $\Theta(n)$ | $8\times10^5$ | 0.00514 s | 0.97 (1) |
| T1 | $\Theta(F)$ | $\Theta(F)$ | $1.5\times10^6$ | 0.0119 s | 1.08 (1) |
| PHD | $\mathcal{O}(n_s\log n_s)$ typical; $\mathcal{O}(n_s^3)$ worst | $\Theta(n_s)$; $\mathcal{O}(n_s^2)$ worst | $\sim5\times10^6$ | 0.117 s† | 0.92 (1+) |

| Step | Time | Space | Ops | Measured | exp. (pred.) |
|---|---|---|---|---|---|
| CRS-A | $\mathcal{O}(F + n \log F)$, $n = 2\times10^5$ | $\Theta(n + N_\varphi)$ | $7\times10^6$ | 0.102 s | 0.97 (1) |
| PAF | $\mathcal{O}(F \log F + F\, N_{\text{fam}})$ | $\Theta(F)$ | $\leq 5\times10^6$ | 0.0892 s‡ | 1.11 (1+) |
| MTS | $\mathcal{O}(F \log F + nc \log F)$ | $\Theta(F + n)$ | $2.6\times10^8$ | 0.03–0.3 s§ | – |

† $H_0$ union–find core only, as a Python loop. ‡ Free-form part with the $N_{\text{fam}} \leq 64$ cap; a prismatic box takes 0.0126 s. § Estimate from Embree throughput; not measured.

## 2.2 Where the time goes for one model

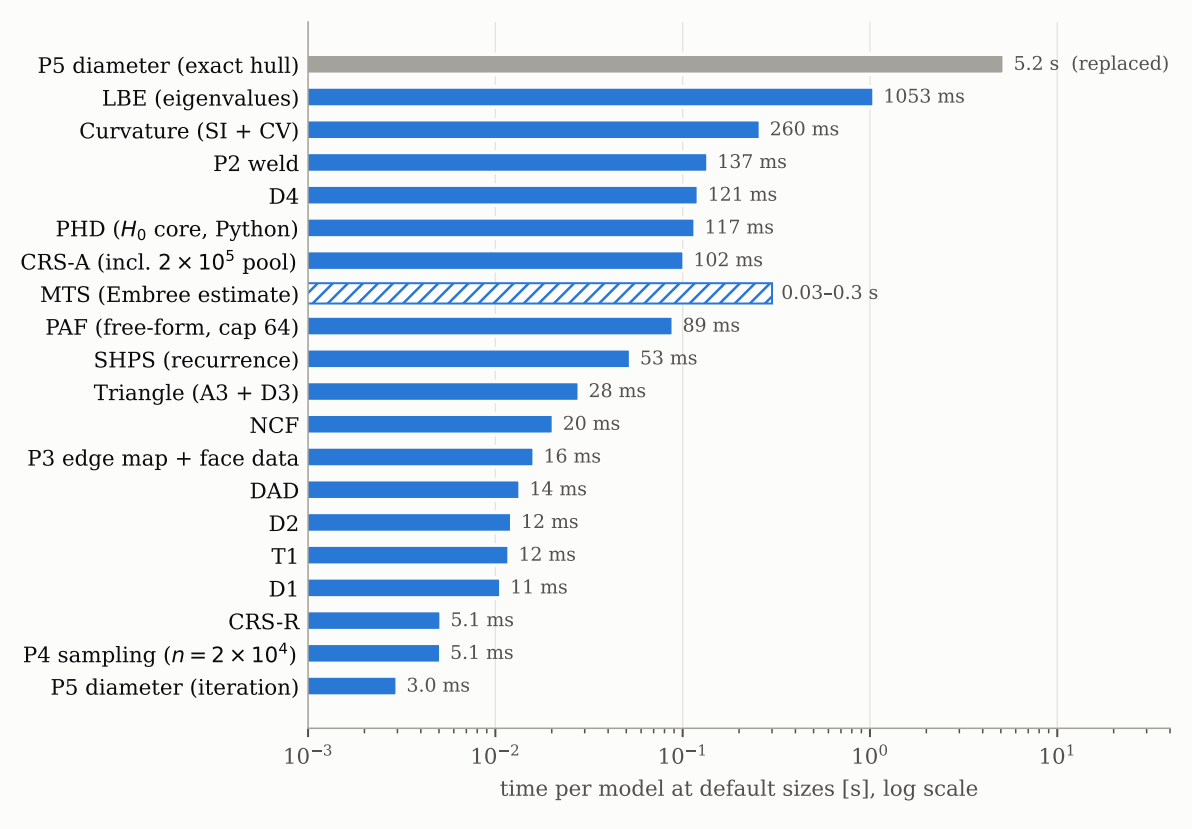



Figure 1: Time per model at the default sizes ($F = 5 \times 10^4$, $n = 2 \times 10^4$, $S = 2 \times 10^5$), measured on the test container (log scale). The hatched bar is an estimate. The grey bar is the exact hull diameter, which the iterative method replaces.

**Totals.** The fifteen descriptors plus preprocessing take about **2.1 s per model** on the test container, or 2.1–2.4 s with MTS on Embree. LBE alone is about half of that (1.05 s), and curvature a further eighth (0.26 s). The remaining thirteen blocks together cost about 0.7 s.

**Peak memory.** The largest vectorised peaks are curvature and D4 at about 77–78 MB. LBE's sparse LU factor has about $4.3 \times 10^6$ non-zeros (≈52 MB), plus a 12 MB Krylov basis. A single worker therefore needs roughly **100 MB**, and under 30 MB if the point-set and curvature blocks are streamed in chunks.

### 2.3 Five findings that matter for implementation

1. **The exact diameter is quadratic on rounded parts.** On curved surfaces a large share of the samples lie on the convex hull ($h = 12165$ at $n = 2 \times 10^4$ here), so comparing all hull pairs is $\mathcal{O}(n^2)$: 5.22 s, more than all fifteen descriptors combined. The double-normal iteration gives the same value to 0.018% in 0.003 s, about 1743× faster.
2. **SHPS must use the Legendre recurrence.** Evaluating each spherical harmonic independently is 27× slower (1.45 s against 0.0528 s) for the same $\Theta(FL^2)$ bound.
3. **LBE dominates and grows fastest with mesh size** (exponent 1.28). Remeshing every part to a fixed vertex budget makes its cost constant per model.
4. **PAF needs its family cap.** Without it, a free-form part turns each facet into a family (11 643 families at $F = 5 \times 10^4$), and the greedy scan becomes $\Theta(F^2)$.
5. **Most descriptors do not depend on mesh size.** D1–D4, NCF, curvature, CRS-R and CRS-A depend only on $n$ and $S$, so a finer STL leaves them unchanged. Mesh size affects only welding, SHPS, DAD, T1, PAF, PHD, LBE and the MTS BVH (§6.3).

## 3 Shared preprocessing

These steps run once per model, and their outputs are reused by several descriptors. Their cost is charged once, not per descriptor.

### 3.1 P1: reading the STL and per-face quantities

An STL stores $F$ triangles, each as 3 corners of 3 coordinates. Reading it is $\Theta(F)$ time and $9F$ words. Areas $A_t = \frac{1}{2}\,\|(\mathbf{x}_1 - \mathbf{x}_0) \times (\mathbf{x}_2 - \mathbf{x}_0)\|$, unit normals and centres each cost $\mathcal{O}(1)$ per face: about 30 flops per face for all three.

**Result**

Time $\Theta(F)$, space $\Theta(F)$ ($7F$ words for areas, normals and centres).

### 3.2 P2: vertex welding

STL corners are not shared, so coincident corners must be merged. Each of the $3F$ corners is quantised to an integer key. With a **hash map** each key is looked up in $\mathcal{O}(1)$ expected time, for $\mathcal{O}(F)$ total. With a **sort** (`np.unique` on the key rows) the cost is $\mathcal{O}(F \log F)$ time and $\Theta(F)$ space.

**Result**

Time $\mathcal{O}(F)$ expected (hash) or $\mathcal{O}(F \log F)$ (sort), space $\Theta(F)$.

**Measured**

Sort-based NumPy weld: exponent 1.09 (predicted 1 to $1 + \epsilon$, since $\log F$ changes little over the range), 0.137 s and 15.2 MB peak at $F = 5 \times 10^4$. This is the most expensive preprocessing step, and it is shared by DAD, LBE, T1 and PHD.

### 3.3 P3: edge–face map

Each face contributes 3 half-edges, $3F$ in total. Sorting their (min, max) vertex keys, or hashing them, groups the two faces of every edge. Edges with one face are boundary edges, and edges with more than two faces are non-manifold.

**Result**

Time $\mathcal{O}(F)$ expected (hash) or $\mathcal{O}(F \log F)$ (sort), space $\Theta(E) = \Theta(F)$.

**Measured**

Exponent 1.14, 0.0161 s and 15.3 MB at $F = 5 \times 10^4$ (together with P1).

### 3.4 P4: area-weighted surface sampling

Build the cumulative area array (prefix sum, $\Theta(F)$). For each of the $n$ samples, draw $u \sim U(0, A)$ and locate its triangle by binary search, $\mathcal{O}(\log F)$, then apply the barycentric formula, $\mathcal{O}(1)$. The Walker–Vose alias table replaces the binary search with an $\mathcal{O}(1)$ lookup after an $\mathcal{O}(F)$ build.

**Result**

Time $\mathcal{O}(F + n \log F)$ (binary search) or $\mathcal{O}(F + n)$ (alias table), space $\Theta(F + n)$.

**Measured**

Against $n$: exponent 0.97, 0.00512 s for $n = 2 \times 10^4$. Against $F$ at fixed $n$: exponent 0.24. The $n \log F$ term dominates, and the prefix sum over $F$ is nearly free.

### 3.5 P5: reference length $D_\varnothing$ (diameter of the sample set)

The companion document computes $D_\varnothing = \max_{i,j} \|\mathbf{p}_i - \mathbf{p}_j\|$ from the convex hull of the pool. The hull costs $\mathcal{O}(n \log n)$ expected (Quickhull). Comparing all pairs of hull vertices then costs $\mathcal{O}(h^2)$. For curved parts a large share of the samples lies on the hull (61% on the test torus, so $h = \Theta(n)$), and this step becomes *quadratic*.

**Result: a bottleneck to avoid**

Exact hull diameter: time $\mathcal{O}(n \log n + h^2)$, which is $\mathcal{O}(n^2)$ for rounded parts; space $\mathcal{O}(n)$ with row chunking.
Double-normal iteration: start at the point farthest from the centre, then repeatedly jump to the point farthest from the current one, $t$ times. Time $\mathcal{O}(tn)$, space $\mathcal{O}(n)$. It returns a lower bound that equals the diameter in practice.
A guaranteed $(1 - \varepsilon)$-approximation takes the largest extent $\max_i \mathbf{u} \cdot \mathbf{p}_i - \min_i \mathbf{u} \cdot \mathbf{p}_i$ over $\mathcal{O}(1/\varepsilon)$ directions $\mathbf{u}$ spread evenly with angular spacing $\sqrt{2\varepsilon}$. Every such extent is at most $D_\varnothing$, and the direction nearest the true diameter gives at least $D_\varnothing \cos\sqrt{2\varepsilon} \geq (1 - \varepsilon) D_\varnothing$. Time $\mathcal{O}(n/\varepsilon)$.

**Measured**

Exact hull diameter: exponent 2.00, 5.22 s at $n = 2 \times 10^4$ ($h = 12165$ hull vertices on a torus). Iteration with $t = 4$: exponent 1.00, 0.003 s at the same $n$, with largest relative error 0.018% over the tested sizes. **The iterative method is about 1743× faster**, and it is the recommended default.

### 3.6 P6: principal axes

The $3 \times 3$ covariance of the pool is one pass, $\Theta(n)$, and its eigen-decomposition is $\mathcal{O}(1)$. CRS-R, PHD and CRS-A share this result.

**Result**

Time $\Theta(n)$, space $\mathcal{O}(1)$ beyond the pool.

### 3.7 P7: spatial index for neighbours (kd-tree)

Building a kd-tree on $n$ points by median splits costs $\mathcal{O}(n \log n)$ time and $\Theta(n)$ space. A $k$-nearest-neighbour query costs $\mathcal{O}(\log n + k)$ expected on well-spread surface samples, so querying every point costs $\mathcal{O}(n(\log n + k))$. Only the curvature block uses it. A fixed-radius query is avoided because the neighbour count, and hence the cost, would depend on the local sampling density.

### 3.8 P8: bounding-volume hierarchy for rays

A surface-area-heuristic BVH is built by binned splitting in $\mathcal{O}(F \log F)$ time and stored in $\Theta(F)$ space (about $2F$ nodes). Only MTS uses it.

## 4 Per-descriptor analysis

Each subsection breaks the algorithm from the companion document into steps and gives the time and space of each step. It then states the total, the operation count at the default sizes, and the measured scaling. The costs of shared preprocessing (§3) are listed as dependencies and not repeated.

### 4.1 D1: radial distance distribution

vector length 32 **Depends on:** P1 (centroid) and P4 (samples).

| Step | Time | Space |
|---|---|---|
| Surface centroid $\mathbf{c} = \sum_t A_t \mathbf{b}_t / A$ | $\Theta(F)$ | $\mathcal{O}(1)$ |
| Distance $\|\mathbf{p}_i - \mathbf{c}\|$ for every sample (8 flops each) | $\Theta(S)$ | $\Theta(S)$ vect., $\mathcal{O}(1)$ streamed |
| Maximum and normalisation | $\Theta(S)$ | in place |
| Histogram (bin index + increment) | $\Theta(S + B)$ | $\Theta(B)$ |

**Result**

Time $\Theta(F + S)$; space $\Theta(S)$ vectorised, $\Theta(B)$ streamed. About $2 \times 10^6$ operations at defaults.

**Measured**

Exponent 0.95 (predicted 1). 0.0107 s and 12.8 MB at $S = 2 \times 10^5$.

### 4.2 D2: pairwise distance distribution

vector length 32 **Depends on:** P4, P5.

| Step | Time | Space |
|---|---|---|
| Draw $S$ index pairs | $\Theta(S)$ | $\Theta(S)$ vect. |
| Difference and norm (12 flops per pair), divide by $D_\varnothing$ | $\Theta(S)$ | $\Theta(S)$ vect., $\mathcal{O}(1)$ streamed |
| Histogram | $\Theta(S + B)$ | $\Theta(B)$ |

**Result**

Time $\Theta(S)$; space $\Theta(S)$ vectorised, $\Theta(B)$ streamed. About $2.4 \times 10^6$ operations at defaults. *All-pairs variant* (every pair of the $n$ pool points): $\Theta(n^2)$ time, $\Theta(B)$ space if processed in row blocks. That is $2 \times 10^8$ pair evaluations for $n = 2 \times 10^4$, about $1000\times$ the default.

**Measured**

Exponent 0.98 (predicted 1). 0.0122 s and 16.0 MB at $S = 2 \times 10^5$.

### 4.3 Triangle descriptor (A3 + D3)

vector length 64 **Depends on:** P4, P5.

| Step | Time | Space |
|---|---|---|
| Draw $S$ index triples; two edge vectors | $\Theta(S)$ | $\Theta(S)$ vect. |
| Cross product, norm, dot product, atan2, square root (about 45 flops per triple) | $\Theta(S)$ | $\Theta(S)$ vect. |
| Two histograms | $\Theta(S + B)$ | $\Theta(B)$ |

**Result**

Time $\Theta(S)$; space $\Theta(S)$ vectorised, $\Theta(B)$ streamed. About $9 \times 10^6$ operations at defaults. The angle and the area share every intermediate, so the two readings cost the same as one.

**Measured**

Exponent 1.07 (predicted 1). 0.0281 s and 30.4 MB at $S = 2 \times 10^5$.

### 4.4 D4: tetrahedron volume with handedness

vector length 33 **Depends on:** P1 (centroid), P4, P5.

| Step | Time | Space |
|---|---|---|
| Draw $S$ index quadruples | $\Theta(S)$ | $\Theta(S)$ vect. |
| Four distances from $\mathbf{c}$ and a sort of 4 values (at most 5 comparisons, a constant) | $\Theta(S)$ | $\Theta(S)$ vect. |
| Triple product (17 flops) and cube root | $\Theta(S)$ | $\Theta(S)$ vect. |
| Histogram; skewness by a one-pass (Welford) moment update | $\Theta(S + B)$ | $\Theta(B)$ |

**Result**

Time $\Theta(S)$; space $\Theta(S)$ vectorised, $\Theta(B)$ streamed. About $1.6 \times 10^7$ operations at defaults. D4 is the most expensive of the point-set descriptors because each quadruple needs a gather of 12 coordinates and the ordering step. Vectorised, the $S \times 4 \times 3$ gathered array alone is 19 MB at $S = 2 \times 10^5$. With the sorted copy, the distances and the index arrays, the measured peak is about $4\times$ that. Streaming in chunks of $10^4$ quadruples keeps it under 4 MB.

**Measured**

Exponent 1.08 (predicted 1). 0.121 s and 76.8 MB at $S = 2 \times 10^5$.

## 4.5 SHPS: orientation spectrum

vector length 15 **Depends on:** P1 (face normals and areas). No sampling is needed.

| Step | Time | Space |
|---|---|---|
| Polar angles $(\theta_t, \phi_t)$ of each normal | $\Theta(F)$ | $\Theta(F)$ |
| Normalised associated Legendre $\bar{P}_\ell^m(\cos\theta_t)$ by the three-term recurrence, $\mathcal{O}(1)$ per $(\ell, m)$ per face; $(L+1)(L+2)/2 = 153$ pairs for $L = 16$ | $\Theta(FL^2)$ | $\Theta(F)$ per pair, or $\Theta(FL)$ per $m$-column |
| $\cos m\phi$, $\sin m\phi$ by the Chebyshev recurrence | $\Theta(FL)$ | $\Theta(F)$ |
| Weighted sums $c_{\ell m} = \sum_t w_t Y_\ell^m(\hat{\mathbf{n}}_t)$ | $\Theta(FL^2)$ | $\Theta(L^2)$ |
| Band powers $\tilde{P}_\ell$ | $\Theta(L^2)$ | $\Theta(L)$ |

**Result**

Time $\Theta(FL^2)$; space $\Theta(F + L^2)$. About $4.6 \times 10^7$ operations at defaults.
*Monte-Carlo variant* using $\tilde{P}_\ell = \mathbb{E}[\mathcal{P}_\ell(\hat{\mathbf{n}}_s \cdot \hat{\mathbf{n}}_t)]$ over $S$ area-weighted face pairs: $\Theta(SL)$ time (one Legendre recurrence per pair) and $\Theta(L)$ space. It is cheaper than the exact form when $S < FL$, at the price of sampling noise.

**Measured**

Recurrence implementation (vectorised over faces, one pass per $(\ell, m)$): exponent 1.10 against $F$ (predicted 1), 0.0528 s and 4.8 MB at $F = 5 \times 10^4$. Against $L = 4..64$ the exponent is 1.57, approaching the predicted 2 at large $L$ (per-call NumPy overhead flattens the small-$L$ end). For comparison, calling SciPy's `sph_harm_y` separately for each $(\ell, m)$ recomputes every Legendre value from scratch: 1.45 s at $F = 5 \times 10^4$, about 27× slower, with $L$-exponent 2.56. **Use the recurrence.**

## 4.6 NCF: normal correlation over distance

vector length 32 **Depends on:** P4 (samples carry their face normals), P5.

| Step | Time | Space |
|---|---|---|
| Draw $S$ pairs; distance and normal dot product (15 flops) | $\Theta(S)$ | $\Theta(S)$ vect., $\mathcal{O}(1)$ streamed |
| Weighted histogram $g(m)$ += $\hat{\mathbf{n}}_i \cdot \hat{\mathbf{n}}_j$ | $\Theta(S + B)$ | $\Theta(B)$ |

**Result**

Time $\Theta(S)$; space $\Theta(S)$ vectorised, $\Theta(B)$ streamed. About $3 \times 10^6$ operations at defaults. If the pairs are shared with D2 (the same index pairs and distances), the extra cost is one dot product per pair.

**Measured**

Exponent 0.99 (predicted 1). 0.0205 s and 16.0 MB at $S = 2 \times 10^5$.

## 4.7 LBE: Laplace–Beltrami eigenvalues

vector length 30 **Depends on:** P2, P3.

| Step | Time | Space |
|---|---|---|
| Cotangent of each triangle corner (3 per face) | $\Theta(F)$ | $\Theta(F)$ |
| Assemble sparse $L$ (about $7V$ non-zeros) and lumped mass $M$ | $\Theta(F)$ hash, $\mathcal{O}(F \log F)$ sort | $\Theta(V)$ |
| Sparse factorisation of $L + \varepsilon M$ (shift-invert). Mesh graphs of bounded genus have $\mathcal{O}(\sqrt{V})$ separators, so nested dissection gives $\mathcal{O}(V \log V)$ fill in $\mathcal{O}(V^{3/2})$ time | $\mathcal{O}(V^{3/2})$ | $\mathcal{O}(V \log V)$ |
| Lanczos (ARPACK), $r$ restarts of $m$ steps. Each step does one forward/back solve $\mathcal{O}(\mathrm{nnz}_{LU})$, one $M$-product $\mathcal{O}(V)$, and re-orthogonalisation $\mathcal{O}(mV)$ | $\mathcal{O}\big(rm(\mathrm{nnz}_{LU} + mV)\big)$ | $\Theta(mV)$ basis |
| Tridiagonal eigenproblem, sorting, logs | $\mathcal{O}(m^2)$ | $\mathcal{O}(m)$ |

**Result**

Time $\mathcal{O}\big(V^{3/2} + r\,m\,(\mathrm{nnz}_{LU} + mV)\big)$; space $\mathcal{O}(\mathrm{nnz}_{LU} + mV)$ with $\mathrm{nnz}_{LU} = \mathcal{O}(V \log V)$ under a nested-dissection ordering.
At defaults ($V = 2.5 \times 10^4$, $m = 61$) the Krylov basis alone is $mV = 1.5 \times 10^6$ words (12 MB), and each restart costs about $10^8$ operations. LBE is the most expensive descriptor in the set.
*Cost levers*: $K$ enters through $m \approx 2K$, so time grows roughly as $K^2$ once re-orthogonalisation dominates. Halving $K$ to 15 cuts the eigen-solve by about 3–4×. Coarsening the mesh to a fixed $V$ (e.g. $10^4$ by remeshing) makes LBE constant-cost per model.

**Measured**

Exponent 1.28 over $V = 6 \times 10^3$–$10^5$ (predicted between 1 and 1.5). 1.05 s at $V = 2.5 \times 10^4$. SciPy's `splu` uses a COLAMD ordering rather than nested dissection. Its measured LU fill was 173 non-zeros per vertex at $V = 2.5 \times 10^4$, growing with exponent 1.22, compared with $\mathcal{O}(\log V)$ per vertex for nested dissection.

### 4.8 DAD: dihedral angle distribution

vector length 38 **Depends on:** P1 (normals, centres), P2, P3.

| Step | Time | Space |
|---|---|---|
| For each of the $E = 1.5F$ edges: two normal gathers, dot product, arccos, convexity sign from the centre difference, and edge length (about 30 flops) | $\Theta(E)$ | $\Theta(E)$ vect., $\mathcal{O}(1)$ streamed |
| Crease test, 36-bin length-weighted histogram, crease length and total bending | $\Theta(E)$ | $\Theta(36)$ |

**Result**

Time $\Theta(F)$; space $\Theta(F)$ vectorised, $\mathcal{O}(1)$ streamed. About $2.3 \times 10^6$ operations at defaults. DAD is dominated by the shared welding and edge map, not by its own work.

**Measured**

Exponent 1.05 (predicted 1). 0.0136 s and 9.1 MB at $F = 5 \times 10^4$ (excluding P2 and P3).

### 4.9 Curvature descriptor (SI + CV)

vector length 97 **Depends on:** P4, P5, P7 (kd-tree).

| Step | Time | Space |
|---|---|---|
| Build the kd-tree on the $n$ samples | $\mathcal{O}(n \log n)$ | $\Theta(n)$ |
| $k$-nearest-neighbour query for every sample | $\mathcal{O}(n(\log n + k))$ | $\Theta(nk)$ indices |
| Local frame; project the $k$ neighbours to $(x, y, z)$ | $\Theta(nk)$ | $\Theta(nk)$ vect. |
| Crease test on the $k$ neighbour normals | $\Theta(nk)$ | $\Theta(nk)$ vect. |
| Normal equations $A^\top A$ ($5 \times 5$) and $A^\top \mathbf{z}$ from the $k \times 5$ design, about $35k$ flops | $\Theta(nk)$ | $\Theta(n)$ |
| Solve $5 \times 5$ ($\approx 45$ flops), $2 \times 2$ eigenvalues, SI, CV | $\Theta(n)$ | $\Theta(n)$ |
| $12 \times 8$ histogram over smooth samples | $\Theta(n)$ | $\Theta(97)$ |

**Result**

Time $\mathcal{O}(n \log n + nk)$; space $\Theta(nk)$ vectorised (the $n \times k \times 5$ design matrix is $2.4 \times 10^7$ bytes at defaults), or $\Theta(n)$ when points are processed in chunks with the tree kept. About $3 \times 10^7$ operations at defaults. This is the second most expensive block after LBE.
A *fixed-radius* neighbourhood would make $k$ depend on the local density: on thin or small features it could reach $\Theta(n)$, giving $\Theta(n^2)$ overall. The $k$-nearest-neighbour form bounds it.

**Measured**

Against $n$: exponent 1.09 (predicted 1 plus a $\log n$ drift), 0.26 s at $n = 2 \times 10^4$. Against $k$: exponent 0.91 (predicted $\leq 1$, since the tree query has a $k$-independent part), and 78.4 MB peak at $k = 30$.

### 4.10 CRS-R: concentric ring profile

vector length 128 **Depends on:** P4, P5, P6.

| Step | Time | Space |
|---|---|---|
| Choose the symmetry axis from the three eigenvalues | $\mathcal{O}(1)$ | $\mathcal{O}(1)$ |
| Three distance-to-axis maps $\|(\mathbf{p} - \mathbf{c}) \times \mathbf{e}_a\|$, about 12 flops each | $\Theta(n)$ | $\Theta(n)$ |
| Histograms with 64 + 32 + 32 bins | $\Theta(n + 128)$ | $\Theta(128)$ |

**Result**

Time $\Theta(n)$; space $\Theta(n)$. About $8 \times 10^5$ operations at defaults.

### 4.11 T1: genus and body count

vector length 4 **Depends on:** P2, P3.

| Step | Time | Space |
|---|---|---|
| Connected components of the face-adjacency graph (BFS, or union–find with inverse-Ackermann cost) | $\Theta(F + E)$ or $\mathcal{O}(E\,\alpha(F))$ | $\Theta(F)$ |
| Per component: count distinct vertices (marking pass), edges ($E_c = \frac{3}{2}F_c + \frac{1}{2}b_c^{\text{edges}}$), faces | $\Theta(F)$ | $\Theta(V)$ marks |
| Trace boundary loops over one-sided edges | $\Theta(\#\text{boundary edges})$ | same |
| Signed volume, 17 flops per face | $\Theta(F)$ | $\mathcal{O}(1)$ per component |

**Result**

Time $\Theta(F)$ (or $\mathcal{O}(F \log F)$ if the vertex count uses sorting); space $\Theta(F)$. About $1.5 \times 10^6$ operations at defaults.

**Measured**

Exponent 1.08 (predicted 1). 0.0119 s and 10.3 MB at $F = 5 \times 10^4$.

## 4.12 PHD: persistent homology

vector length 77 **Depends on:** P2, P3, P6.

| Step | Time | Space |
|---|---|---|
| Filtration value per vertex (radius from the axis); lower-star values for edges and faces | $\Theta(n_s)$ | $\Theta(n_s)$ |
| Sort the $n_s \approx 3F$ simplices by value | $\mathcal{O}(n_s \log n_s)$ | $\Theta(n_s)$ |
| $H_0$ pairs: union–find over edges in sorted order (elder rule) | $\mathcal{O}(E\,\alpha(V))$ | $\Theta(V)$ |
| $H_2$ and the upward $H_1$ pairs of a 2-manifold: $H_0$ of $-f$ on the dual graph (duality) | $\mathcal{O}(E\,\alpha(F))$ | $\Theta(F)$ |
| Extended persistence: pairing of the $2g$ essential loops. General boundary-matrix reduction on the coned complex ($n_s' \approx 2n_s$) | worst $\mathcal{O}(n_s^3)$; near-linear in practice with clearing and twist | worst $\mathcal{O}(n_s^2)$, typical $\Theta(n_s)$ |
| Three persistence images on a $5 \times 5$ grid | $\mathcal{O}(\lvert\mathrm{PD}\rvert\, R^2)$ | $\Theta(R^2)$ |

**Result**

Time $\mathcal{O}(n_s \log n_s)$ typical, $\mathcal{O}(n_s^3)$ worst case; space $\Theta(n_s)$ typical, $\mathcal{O}(n_s^2)$ worst. At defaults $n_s \approx 1.5 \times 10^5$, so the typical cost is a few $10^6$ operations. Worst-case behaviour is not expected on mesh filtrations, but it is the reason PHD is ranked last for adoption.

**Measured**

The $H_0$ union–find core, written as a plain Python loop, has exponent 0.92 (predicted $1 + \epsilon$ from the sort) and takes 0.117 s at $F = 5 \times 10^4$. A compiled library such as GUDHI runs the same step tens of times faster. The full extended persistence was not run (GUDHI is not available in the sandbox).

## 4.13 CRS-A: angular pattern spectrum

vector length 128 **Depends on:** P4 with its own, larger pool ($n = 2 \times 10^5$), P6.

| Step | Time | Space |
|---|---|---|
| Extra samples: the pool is $10\times$ larger than for the other blocks | $\mathcal{O}(F + n \log F)$ | $\Theta(n)$ |
| Azimuth $\varphi = \mathrm{atan2}(\cdot, \cdot)$ and radial band for each sample | $\Theta(n)$ | $\Theta(n)$ |
| Four band histograms with $N_\varphi = 256$ bins | $\Theta(n + 4N_\varphi)$ | $\Theta(N_\varphi)$ |
| Four real FFTs of length 256 | $\Theta(4N_\varphi \log N_\varphi) \approx 8 \times 10^3$ | $\Theta(N_\varphi)$ |

**Result**

Time $\mathcal{O}(F + n \log F)$; space $\Theta(n + N_\varphi)$. About $7 \times 10^6$ operations at defaults, almost all of it drawing the larger pool. The FFTs are negligible.

**Measured**

CRS-R and CRS-A together: exponent 0.97 (predicted 1). 0.00514 s and 1.8 MB at $n = 2 \times 10^4$; 0.0516 s at $n = 2 \times 10^5$.

### 4.14 PAF: planar face families

vector length 31 **Depends on:** P1.

| Step | Time | Space |
|---|---|---|
| Second- and fourth-order normal moments $T_2$ (6 products) and $T_4$ (15 products) | $\Theta(21F)$ | $\mathcal{O}(1)$ |
| Sort faces by area | $\mathcal{O}(F \log F)$ | $\Theta(F)$ |
| Greedy family detection: each new family scans the remaining faces | $\mathcal{O}(F\, N_{\text{fam}})$ | $\Theta(F)$ |
| Offsets per family: dot product, sort, 1-D gap clustering | $\mathcal{O}(F \log F)$ total | $\Theta(F)$ |
| Normal entropy on the icosphere: nearest of $G = 2562$ cells via a kd-tree | $\mathcal{O}(F \log G)$ | $\Theta(G)$ |

**Result**

Time $\mathcal{O}(F \log F + F\, N_{\text{fam}})$; space $\Theta(F)$.
On a prismatic part $N_{\text{fam}}$ is small (3 for a box, up to about 20 for a machined housing), and each scan removes a large share of the faces, so the cost is close to $\mathcal{O}(F \log F)$. On a free-form part every facet can form its own "family", making $N_{\text{fam}} = \Theta(F)$ and the scan $\Theta(F^2)$. The **cap** $N_{\text{fam}} \le 64$ (remaining faces are labelled non-planar) bounds this at $\mathcal{O}(64F)$. The cap is necessary for correctness of the cost bound, not only for speed.

**Measured**

Box meshes ($N_{\text{fam}} = 3$): exponent 0.99, 0.0126 s at $F \approx 50\,700$. Free-form torus with the cap: exponent 1.11, 0.0892 s at $F = 5 \times 10^4$. Without the cap the torus produced 11 643 families at $F = 5 \times 10^4$.

### 4.15 MTS: wall thickness spectrum

vector length 40 **Depends on:** P4, P8 (BVH).

| Step | Time | Space |
|---|---|---|
| Cone of $c = 16$ directions per sample (fixed pattern, rotated to the normal) | $\Theta(nc)$ | $\mathcal{O}(c)$ |
| First-hit query for each of the $nc$ rays: $\mathcal{O}(\log F)$ expected node visits on a balanced BVH, $\Theta(F)$ in the worst case (rays grazing many coplanar triangles) | $\mathcal{O}(nc \log F)$ expected | $\mathcal{O}(\log F)$ stack |
| Robust median per sample (selection on $c$ values) | $\Theta(nc)$ | $\mathcal{O}(c)$ |
| Log histogram and moments | $\Theta(n + B)$ | $\Theta(B)$ |

**Result**

Time $\mathcal{O}(F \log F + nc \log F)$ expected, including the BVH build; $\mathcal{O}(ncF)$ worst case. Space $\Theta(F)$ for the BVH plus $\Theta(n)$ for thicknesses. At defaults there are $nc = 3.2 \times 10^5$ rays with about 16 node visits each: roughly $2.6 \times 10^8$ operations.
*Wall-clock depends almost entirely on the ray tracer*: Embree reaches $10^6$–$10^7$ rays/s per core ($\approx$ 0.03–0.3 s here), while a pure-Python ray–triangle loop manages $10^4$–$10^5$ rays/s ($\approx$ 3–30 s).

*Cost levers*: $c = 8$ rays halves the cost with a small loss of robustness; the thickness histogram needs far fewer samples than $n = 2 \times 10^4$, and $n = 5 \times 10^3$ is usually enough.

# 5 Assembly, compression and search

## 5.1 Building the final vector (per model)

| Step | Time | Space |
|---|---|---|
| Hellinger map $\sqrt{\mathbf{h}_b}$ on every histogram block | $\Theta(d)$ | $\Theta(d)$ |
| Standardise the scalar blocks with the stored database mean and std | $\Theta(d)$ | $\Theta(d)$ stored statistics |
| Scale each block to unit length and multiply by $\sqrt{\beta_b}$ | $\Theta(d)$ | $\Theta(d)$ |

**Result**

Time and space $\Theta(d)$ with $d = 781$. This is negligible next to the descriptors.

## 5.2 PCA compression (offline, once per database)

For a block of length $d_b$ and $M$ models, the covariance costs $\Theta(Md_b^2)$, its eigen-decomposition $\Theta(d_b^3)$, and the stored basis $\Theta(d_b k_b)$. Compressing **per block** rather than on the whole 781-long vector changes the cost from $\Theta(Md^2 + d^3)$ to $\Theta(M\sum_b d_b^2 + \sum_b d_b^3)$. With the block lengths of the companion document:

$$\sum_b d_b^2 = 6.2 \times 10^4 \text{ vs. } d^2 = 6.1 \times 10^5, \qquad \sum_b d_b^3 = 6.1 \times 10^6 \text{ vs. } d^3 = 4.8 \times 10^8.$$

Per-block PCA is therefore about $10\times$ cheaper in the covariance and $78\times$ cheaper in the eigen-decomposition. It is also better conditioned, since every block has $d_b \ll M$. At $M = 600$ the whole fit is about $4 \times 10^7$ operations, under a second.

**Result**

Offline fit: time $\Theta(M\sum_b d_b^2 + \sum_b d_b^3)$, space $\Theta(Md + \sum_b d_b^2)$.
Projecting one query: time $\Theta(\sum_b d_b k_b) \le \Theta(dk)$, space $\Theta(k)$.

## 5.3 Nearest-neighbour search

| Index | Query time | Memory | Notes |
|---|---|---|---|
| Flat inner product | $\Theta(Mk)$ | $4Mk$ bytes | exact; batched queries use matrix–matrix products |
| IVF ($n_\ell$ lists, $n_p$ probed) | $\mathcal{O}(n_\ell k + \frac{n_p}{n_\ell} Mk)$ | $4Mk$ bytes | $n_\ell \approx \sqrt{M}$; approximate |
| IVF-PQ ($m_q$ sub-quantisers) | $\mathcal{O}(n_\ell k + \frac{n_p}{n_\ell} Mm_q)$ | $\approx Mm_q$ bytes | compressed codes; approximate |
| HNSW ($R$ links, beam $e$) | $\approx \mathcal{O}(e\,k \log M)$ | $4Mk + 8MR$ bytes | best CPU latency and recall |

| Database size $M$ | 600 | $10^4$ | $10^5$ | $10^6$ |
|---|---|---|---|---|
| Flat query, $k = 100$ (multiply-adds) | $6 \times 10^4$ | $10^6$ | $10^7$ | $10^8$ |
| Flat query, approximate time | $\mu$s | $< 1$ ms | $\approx 1$ ms | 5–20 ms |
| Index memory, $k = 100$, float32 | 0.24 MB | 4 MB | 40 MB | 400 MB |
| Raw 781-long features, float32 | 1.9 MB | 31 MB | 312 MB | 3.1 GB |
| Offline extraction at 2.1 s/model, 1 core | 21 min | 5.8 h | 58 h | 24 days |

**Reading the table.** Search is never the bottleneck at the current corpus size: a flat index answers in microseconds. *Feature extraction* is what scales with $M$. It is embarrassingly parallel across models, so wall-clock time divides by the number of cores or machines. Each model needs to be processed only once and can then be cached, which `cache.py` in the codebase already supports.

### 5.4 Per query

A query CAD model goes through the same extraction (about 2.1 s here, dominated by LBE), one projection ($\Theta(dk)$), and one search ($\Theta(Mk)$ flat). Query latency is therefore set by extraction, and specifically by LBE and curvature. If faster queries are required, remeshing to a fixed vertex budget for LBE, and processing curvature in chunks on fewer samples ($n = 10^4$), bring extraction to roughly 1 s.

## 6 Measured scaling

## 6.1 Scaling curves

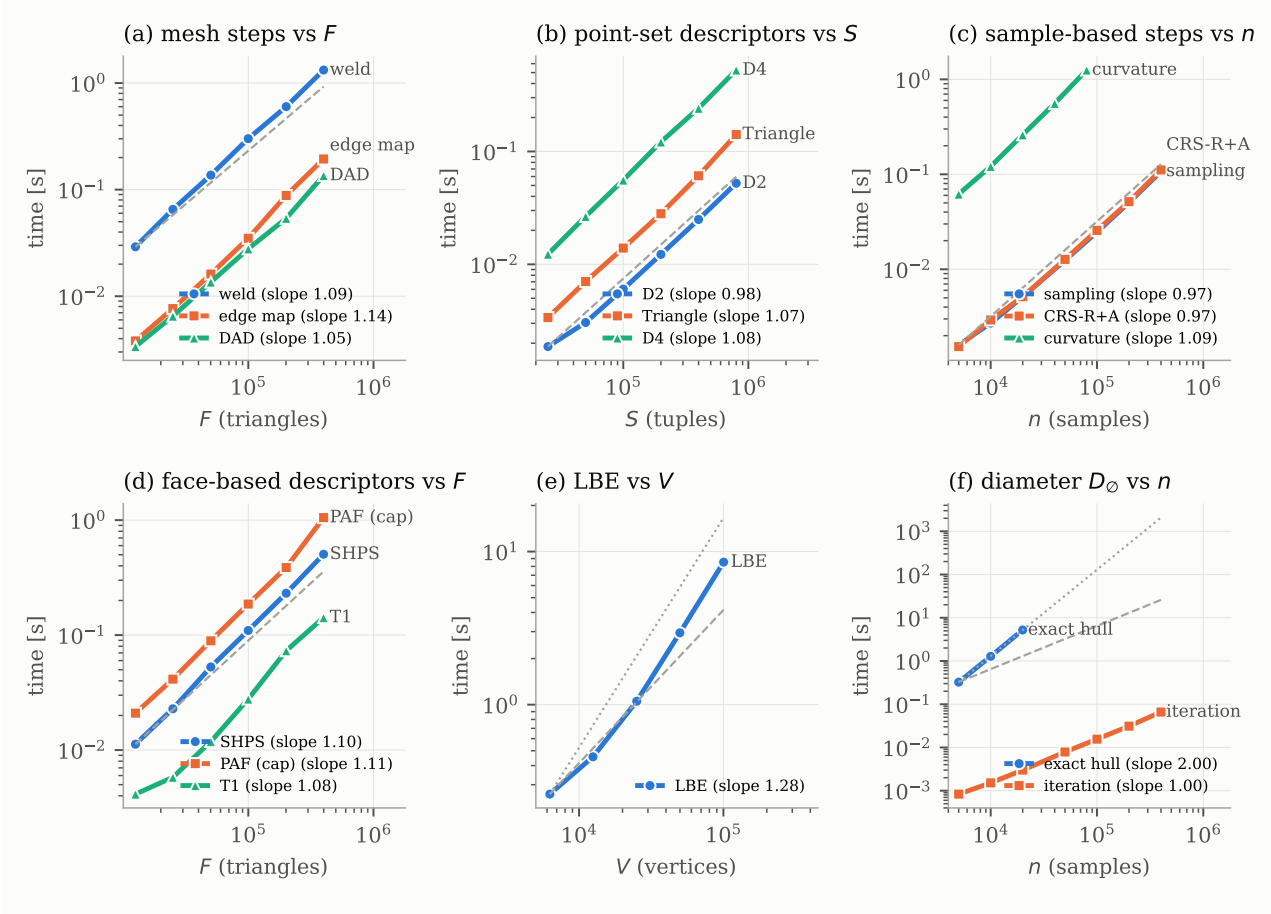



Figure 2: Measured time against input size (log–log), with the fitted slope in each legend. Grey dashed lines have slope 1. In (e) the dotted line has slope 1.5, and in (f) slope 2. A straight line of slope $p$ means the cost grows as size$^p$.

## 6.2 Predicted against measured exponents

| Step | Varied | Predicted | Measured | Comment |
|---|---|---|---|---|
| P2 weld | $F$ | $1(+\log)$ | 1.09 | sort-based; the $\log F$ factor shows as $+0.09$ |
| P3 edge map | $F$ | $1(+\log)$ | 1.14 | sort-based |
| P4 sampling | $n$ | 1 | 0.97 | |
| P4 sampling | $F$ | $\approx 0$ | 0.24 | $n \log F$ dominates; prefix sum is cheap |
| P5 exact diameter | $n$ | 2 | 2.00 | hull holds a constant share of points |
| P5 iteration | $n$ | 1 | 1.00 | |
| D1, D2, NCF | $S$ | 1 | 0.95, 0.98, 0.99 | |
| Triangle, D4 | $S$ | 1 | 1.07, 1.08 | slight excess from cache misses on large gathers |
| SHPS | $F$ | 1 | 1.10 | recurrence form |
| SHPS | $L$ | 2 | 1.57 | per-call overhead at small $L$; local slope 1.8 from $L = 32$ to 64 |
| LBE | $V$ | 1–1.5 | 1.28 | COLAMD fill grows as $V^{1.22}$ |
| DAD | $F$ | 1 | 1.05 | |
| Curvature | $n$ | $1(+\log)$ | 1.09 | kd-tree $\log n$ term |
| Curvature | $k$ | $\leq 1$ | 0.91 | tree traversal is independent of $k$ |
| CRS-R + CRS-A | $n$ | 1 | 0.97 | |
| T1 | $F$ | 1 | 1.08 | |
| PHD ($H_0$) | $F$ | $1(+\log)$ | 0.92 | Python loop; fixed overhead at small $F$ |
| PAF (box) | $F$ | $1(+\log)$ | 0.99 | 3 families |
| PAF (free-form, cap) | $F$ | $1(+\log)$ | 1.11 | 64 scans |

Every measured exponent agrees with its prediction to within the log factors and fixed overheads noted. No step shows unexpected super-linear growth.

## 6.3 Effect of mesh size

The table splits the per-model time into the part that depends on the mesh (welding, edge map, SHPS, DAD, T1, PAF, PHD core) and LBE, measured on torus meshes. The sample-based part (D1–D4, NCF, curvature, CRS, sampling and diameter) stays at about 0.57 s whatever the mesh size, because it is fixed by $n$ and $S$.

| $F$ | mesh-dependent part | LBE | sample-based part | total per model |
|---|---|---|---|---|
| 12 544 | 0.12 s | 0.26 s | 0.57 s | 0.95 s |
| 24 964 | 0.21 s | 0.46 s | 0.57 s | 1.23 s |
| 50 176 | 0.44 s | 1.05 s | 0.57 s | 2.06 s |
| 99 856 | 0.97 s | 2.95 s | 0.57 s | 4.49 s |
| 200 704 | 1.91 s | 8.53 s | 0.57 s | 11.0 s |

LBE's share rises from 27% to 78% as the mesh grows sixteen-fold. Its super-linear growth is the only reason total cost grows faster than the mesh. Remeshing to a fixed budget of $V \approx 10^4$ vertices before LBE caps it at about 0.4 s. The spectrum's low eigenvalues barely change, because they describe coarse shape.

## 6.4 Reducing cost: ranked by payoff

1. **Diameter by iteration instead of exact hull pairs**: removes a quadratic step (1743× faster at defaults).

2. **Fixed vertex budget for LBE** (isotropic remeshing, or decimation, to $V \approx 10^4$): makes the largest block constant-cost. Also use a nested-dissection ordering (METIS/CHOLMOD) instead of SciPy's COLAMD to reduce LU fill (measured 173 non-zeros per vertex).
3. **SHPS by recurrence**: $27\times$ faster than per-harmonic evaluation.
4. **Share samples and intermediates.** D2 and NCF use the same pairs; the triangle and D4 blocks can draw from the same index arrays; one pool and one set of principal axes serve D1–D4, NCF, curvature and CRS-R. Only CRS-A needs its own larger pool.
5. **Stream in chunks** of $10^4$ tuples or points for D4 and curvature. Peak memory drops from about 78 MB to a few MB at no cost in time.
6. **Cap PAF families** at 64. This bounds the cost on free-form parts, where it would otherwise be quadratic.
7. **Use Embree for MTS**, and 8 rays per sample instead of 16 if more speed is needed.
8. **Parallelise and cache.** Extraction is independent per model and, after preprocessing, per descriptor. With $P$ cores the database build takes $\approx 2.1M/P$ s. Features are computed once per model version and cached.